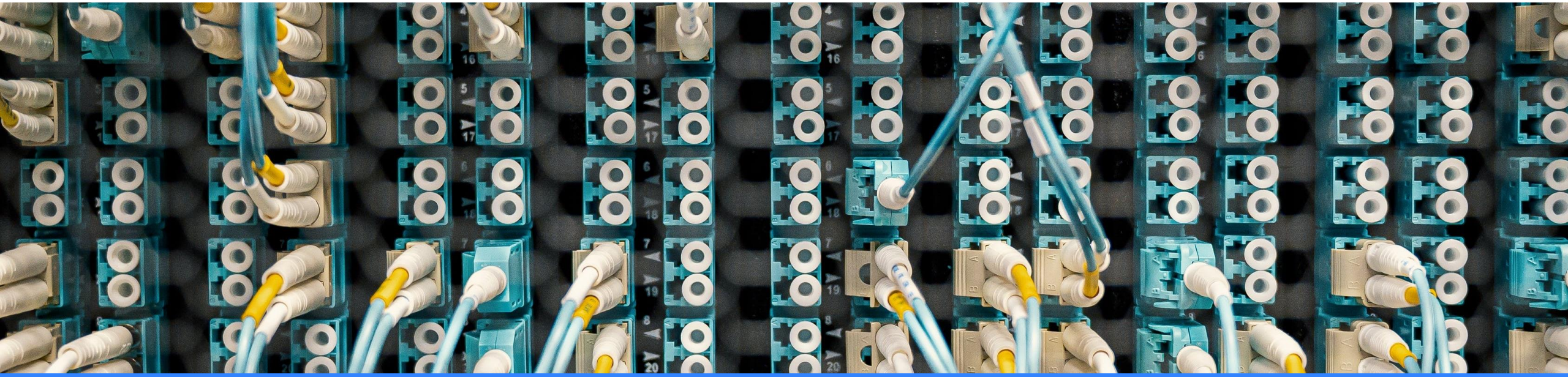


Artificial Intelligence in Interactive Systems

Recurrent Neural Network and Long Short-Term Memory

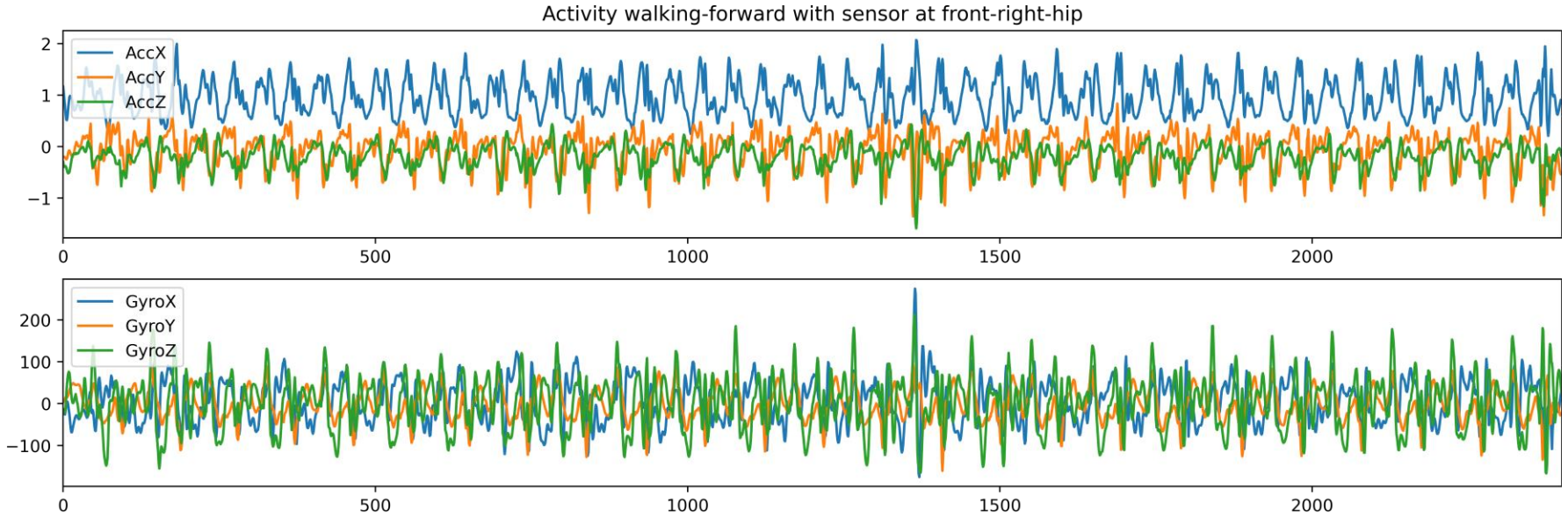
Sven Mayer

Model Layers

- Dense Layer
  - No specific relationship between inputs
- CNN Layer
  - Budling an understanding by looking at neighboring inputs in an n-dimensional input

Sequence Data

- Time Series Data
- Natural Language

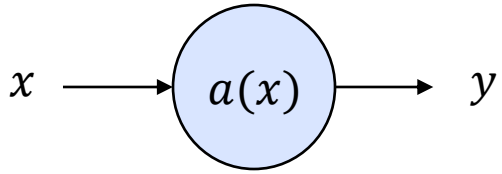


Data Source: https://dl.acm.org/doi/10.1145/2370216.2370438

Recurrent Neural Network (RNN)

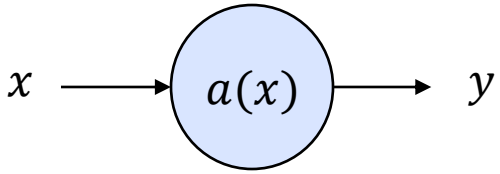
Recap Perceptron

Perceptron

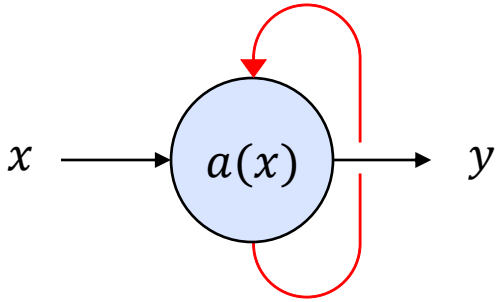


Recurrent Neural Network (RNN)

Perceptron



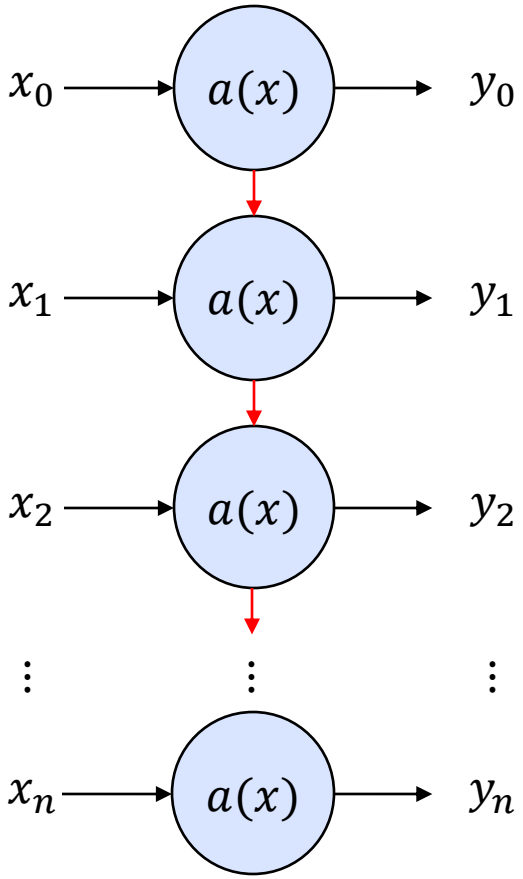
RNN Unit

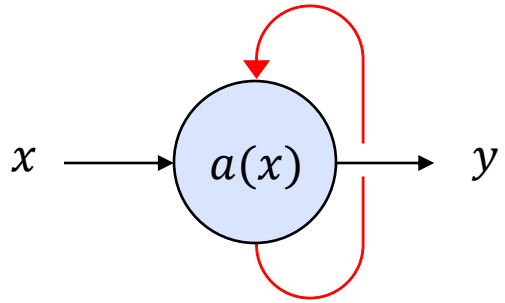


Hidden State

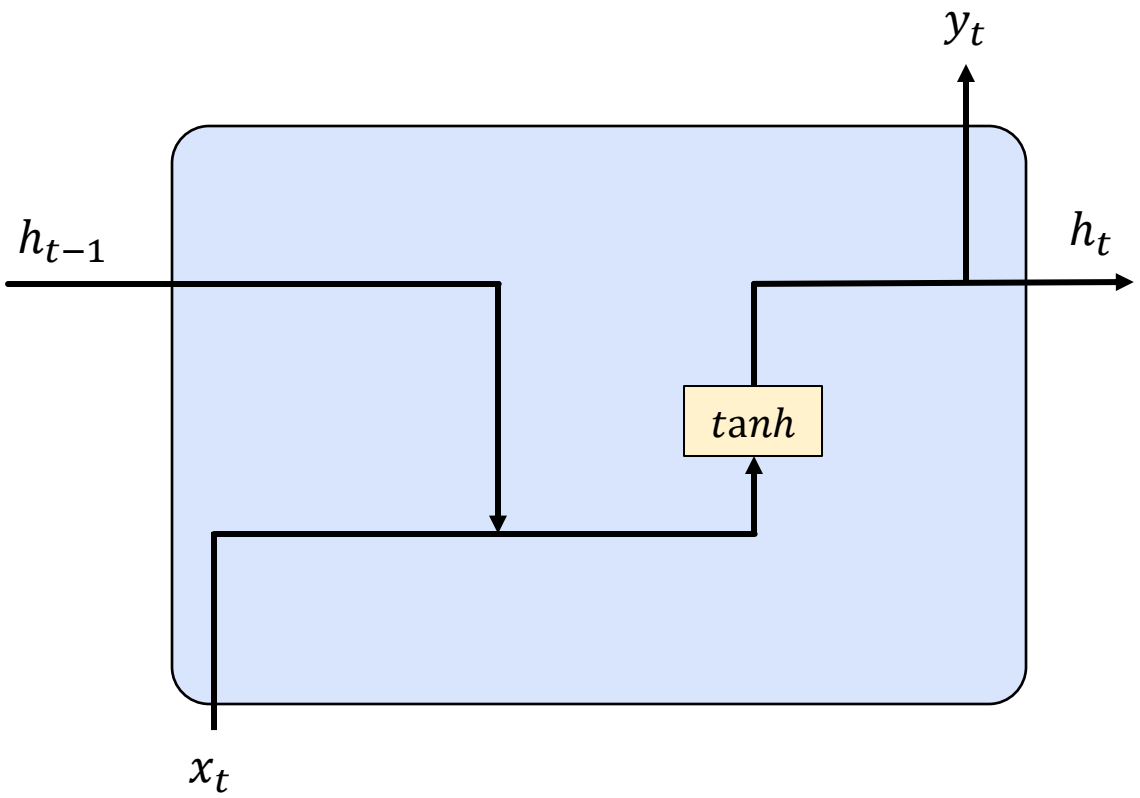
Recurrent Neural Network

Unrolled Recurrent Neural Network

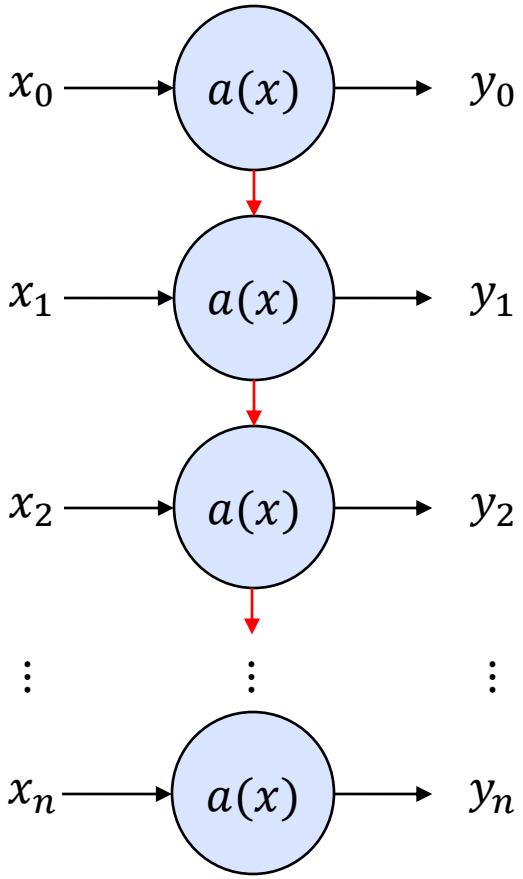




Recurrent Neural Network



Recurrent Neural Network



- Each state gets the “recent” information
- Long-term dependencies gets lost

RNN in TensorFlow

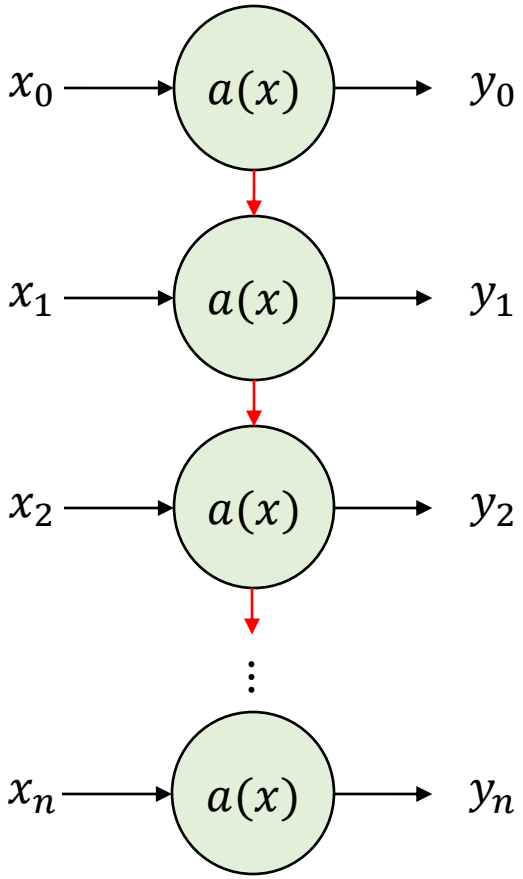
Two RNN layer using a fixed window length

```
inputs = Input(shape=(WINDOW_LENGTH, 3), name="Input")
x = SimpleRNN(32, name="RNN_1", return_sequences=True)(inputs)
x = SimpleRNN(32, name="RNN_2")(x)
prediction = Dense(len(classes), activation="softmax", name="Output")(x)
model_functional = tf.keras.Model(inputs = inputs, outputs = prediction)
model_functional.summary()
```

```
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
Input (InputLayer)           [(None, 200, 3)]          0
_________________________________________________________________
RNN_1 (SimpleRNN)            (None, 200, 32)           1152
_________________________________________________________________
RNN_2 (SimpleRNN)            (None, 32)                2080
_________________________________________________________________
Output (Dense)               (None, 12)                396
=================================================================
Total params: 3,628
Trainable params: 3,628
Non-trainable params: 0
```

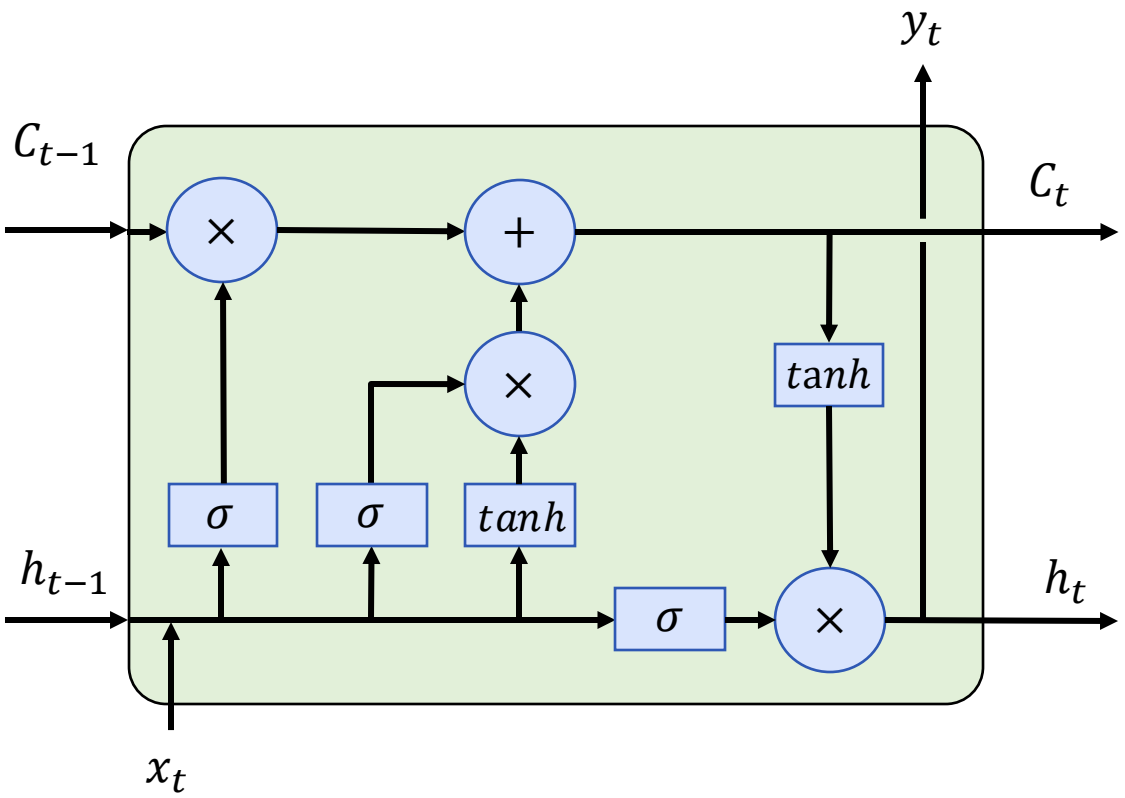
Long Short-Term Memory (LSTM)

Long Short-Term Memory (LSTM)



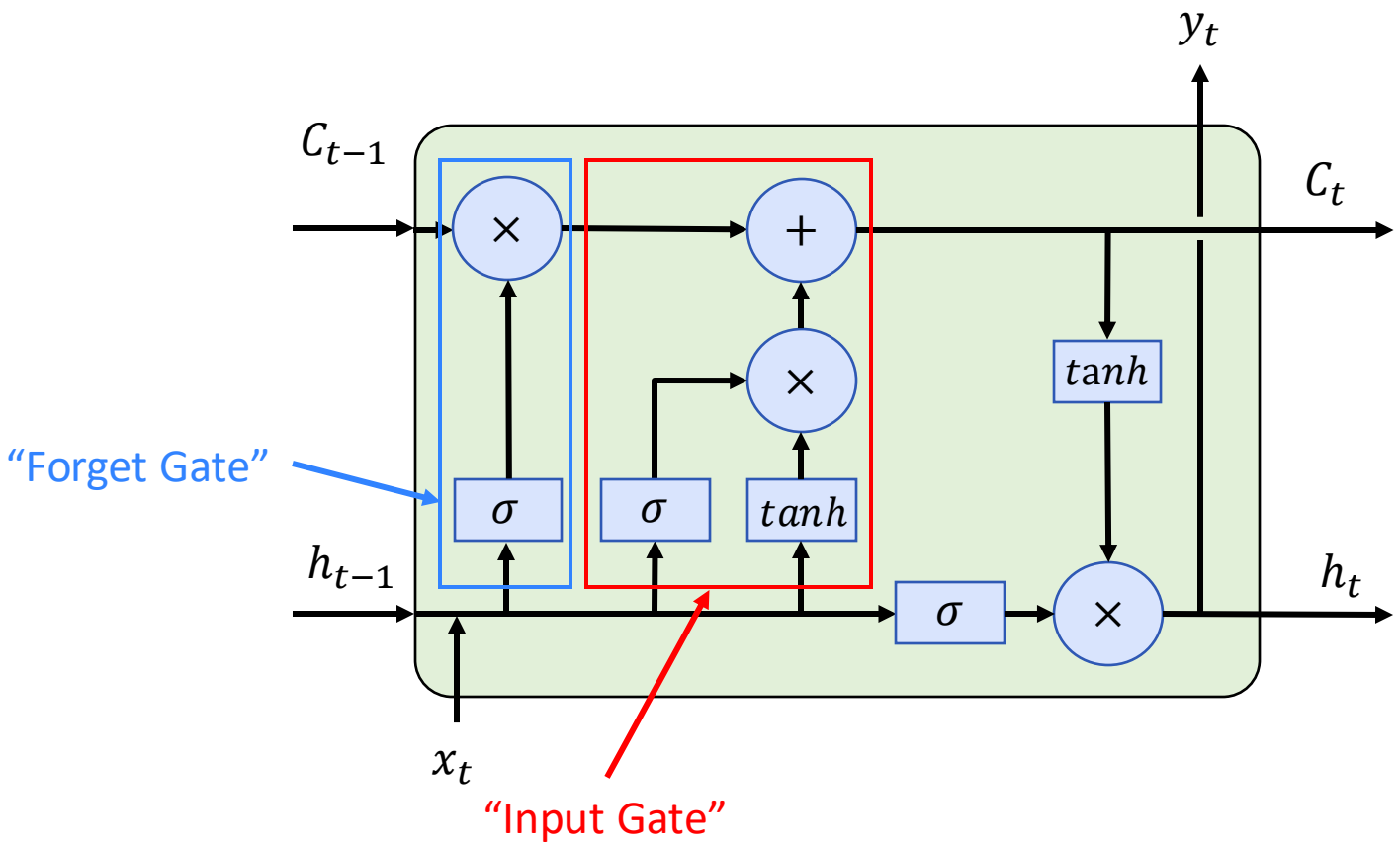
Sepp Hochreiter, and Schmidhuber Jürgen. "Long short-term memory." *Neural computation* (1997).

Long Short-Term Memory (LSTM)



Credit: http://colah.github.io/posts/2015-08-Understanding-LSTMs/

Long Short-Term Memory (LSTM)



LSTM in TensorFlow

Two RNN layer using a fixed window length

```
inputs = Input(shape=(WINDOW_LENGTH, 3), name="Input")
x = LSTM(32, name="LSTM_1", return_sequences=True)(inputs)
x = LSTM(32, name="LSTM_2")(x)
prediction = Dense(len(classes), activation="softmax", name="Output")(x)
model_functional = tf.keras.Model(inputs = inputs, outputs = prediction)
model_functional.summary()
```

```
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
Input (InputLayer)           [(None, 200, 3)]          0
_________________________________________________________________
LSTM_1 (LSTM)                (None, 200, 32)           4608
_________________________________________________________________
LSTM_2 (LSTM)                (None, 32)                8320
_________________________________________________________________
Output (Dense)               (None, 12)                396
=================================================================
Total params: 13,324
Trainable params: 13,324
Non-trainable params: 0
```

Conclusion

Recurrent Neural Network and Long Short-Term Memory

- Processing sequence data using neuronal networks
- Recurrent Neural Network (RNN)
- Long Short-Term Memory (LSTM)

- Variations
  - Gated Recurrent Unit (GRU) [1]

[1] Cho, Kyunghyun, et al. "Learning phrase representations using RNN encoder-decoder for statistical machine translation." *arXiv preprint arXiv:1406.1078* (2014).

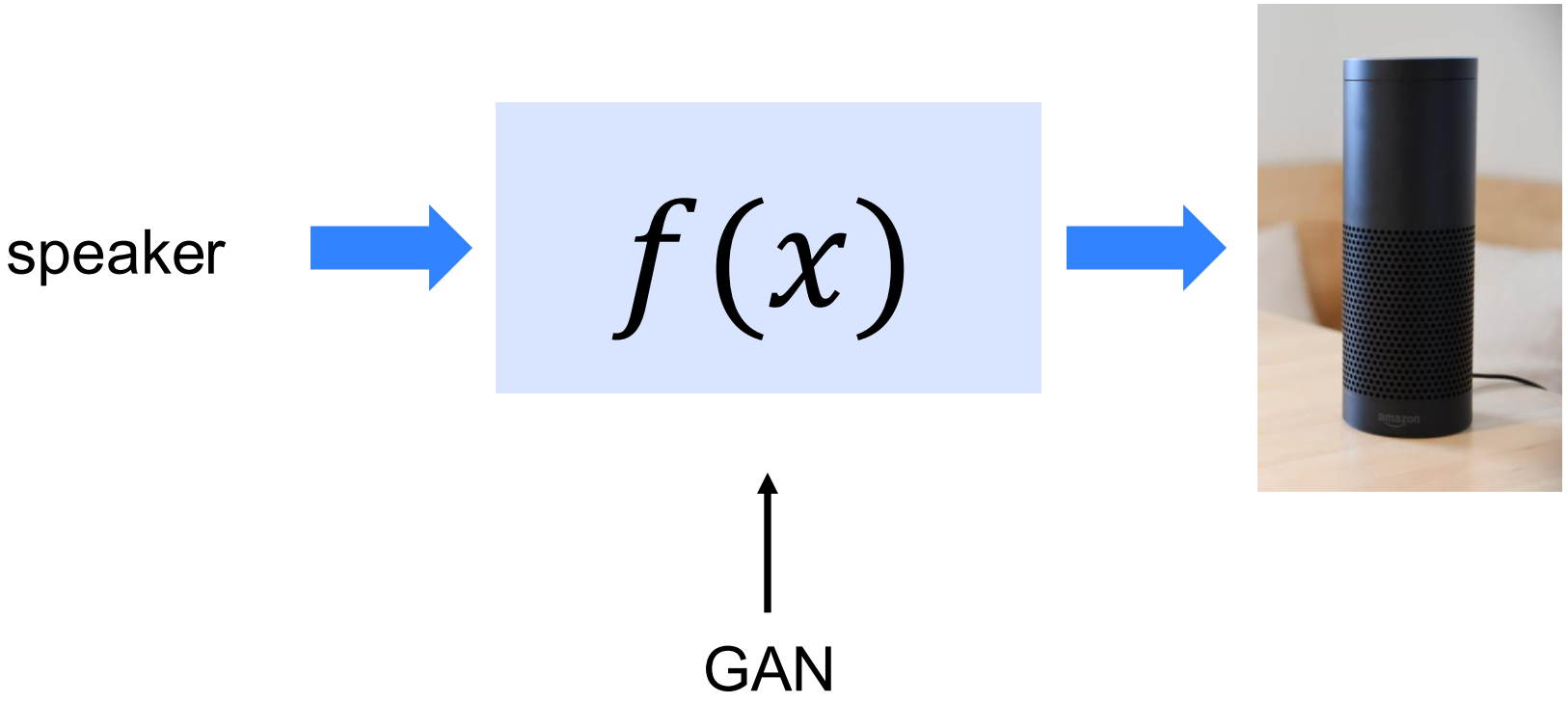
License

Artificial Intelligence in Interactive Systems

Generative Adversarial Network

Sven Mayer

Generative Adversarial Network (GAN)



Face Generation



Source: https://thispersondoesnotexist.com/

Karras, Tero, et al. "Analyzing and improving the image quality of stylegan." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2020. DOI: https://arxiv.org/abs/1912.04958

Style Transfer

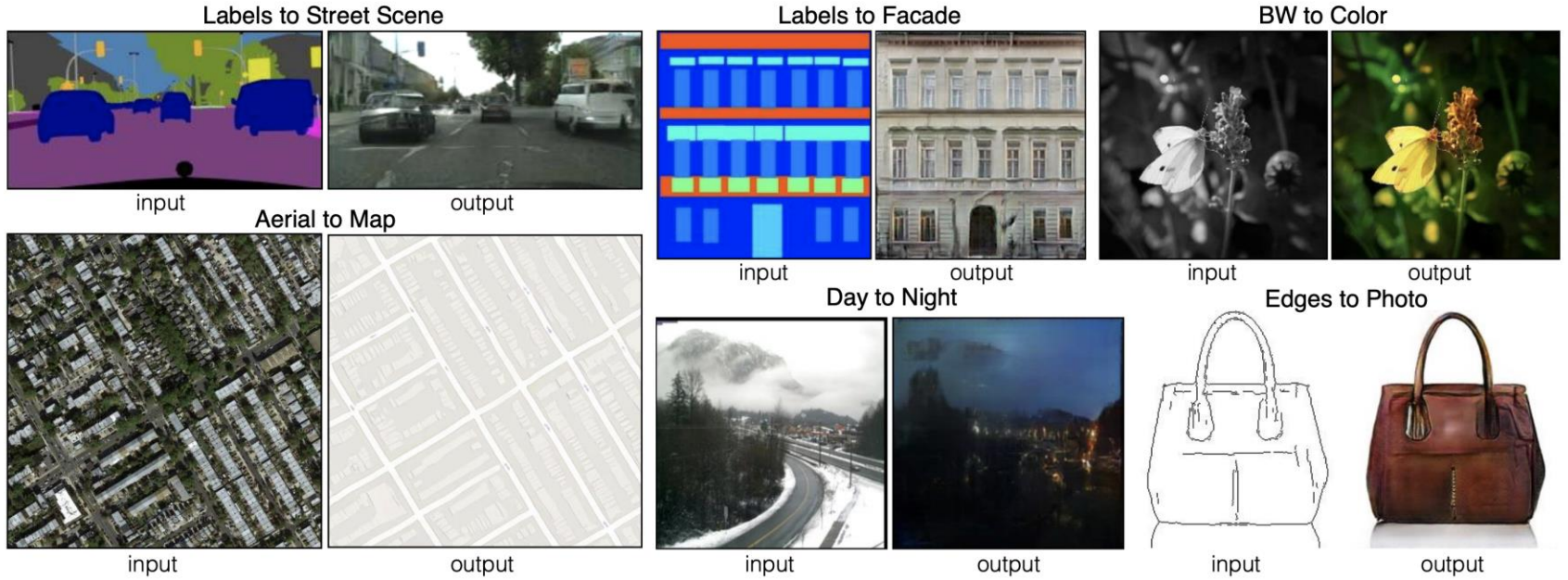


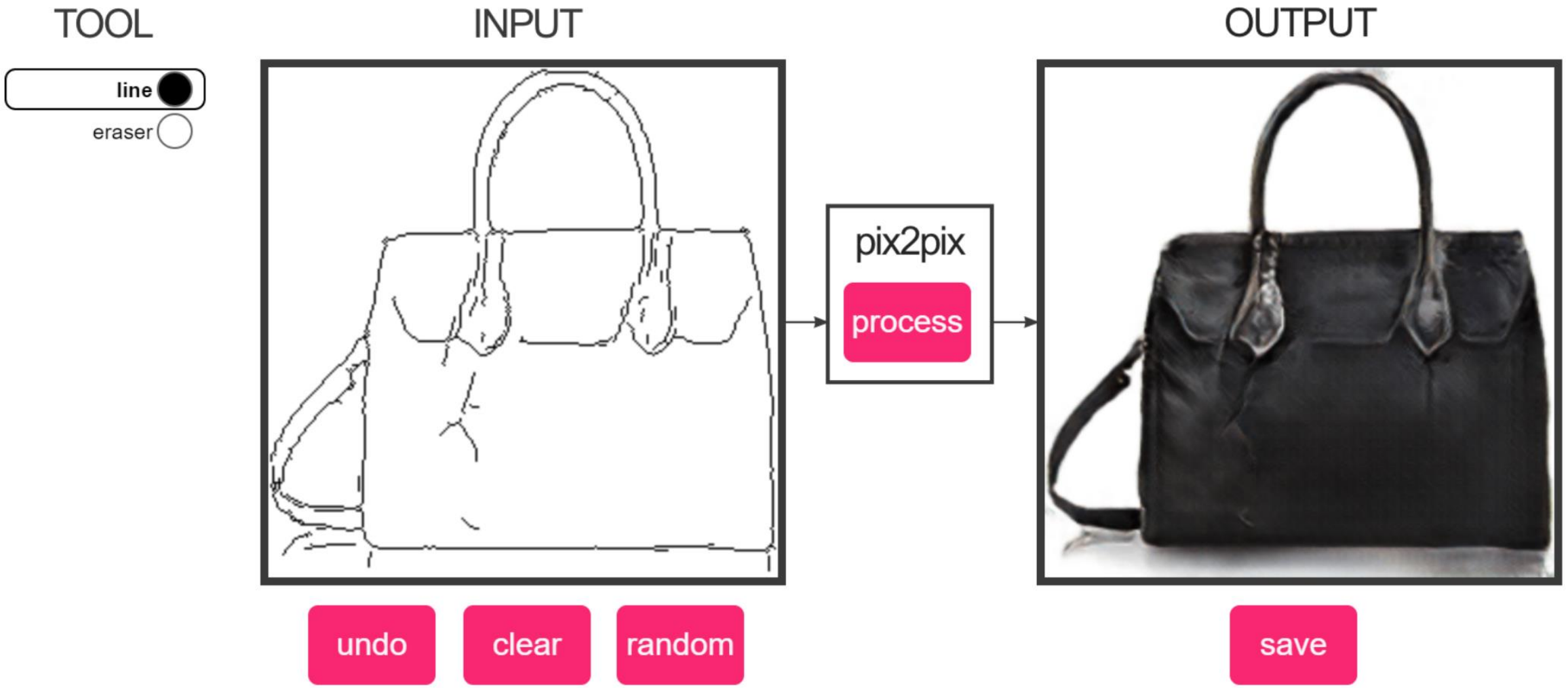
Figure 1: Many problems in image processing, graphics, and vision involve translating an input image into a corresponding output image. These problems are often treated with application-specific algorithms, even though the setting is always the same: map pixels to pixels. Conditional adversarial nets are a general-purpose solution that appears to work well on a wide variety of these problems. Here we show results of the method on several. In each case we use the same architecture and objective, and simply train on different data.

Isola, Phillip, et al. "Image-to-image translation with conditional adversarial networks." *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2017. DOI: https://arxiv.org/abs/1611.07004

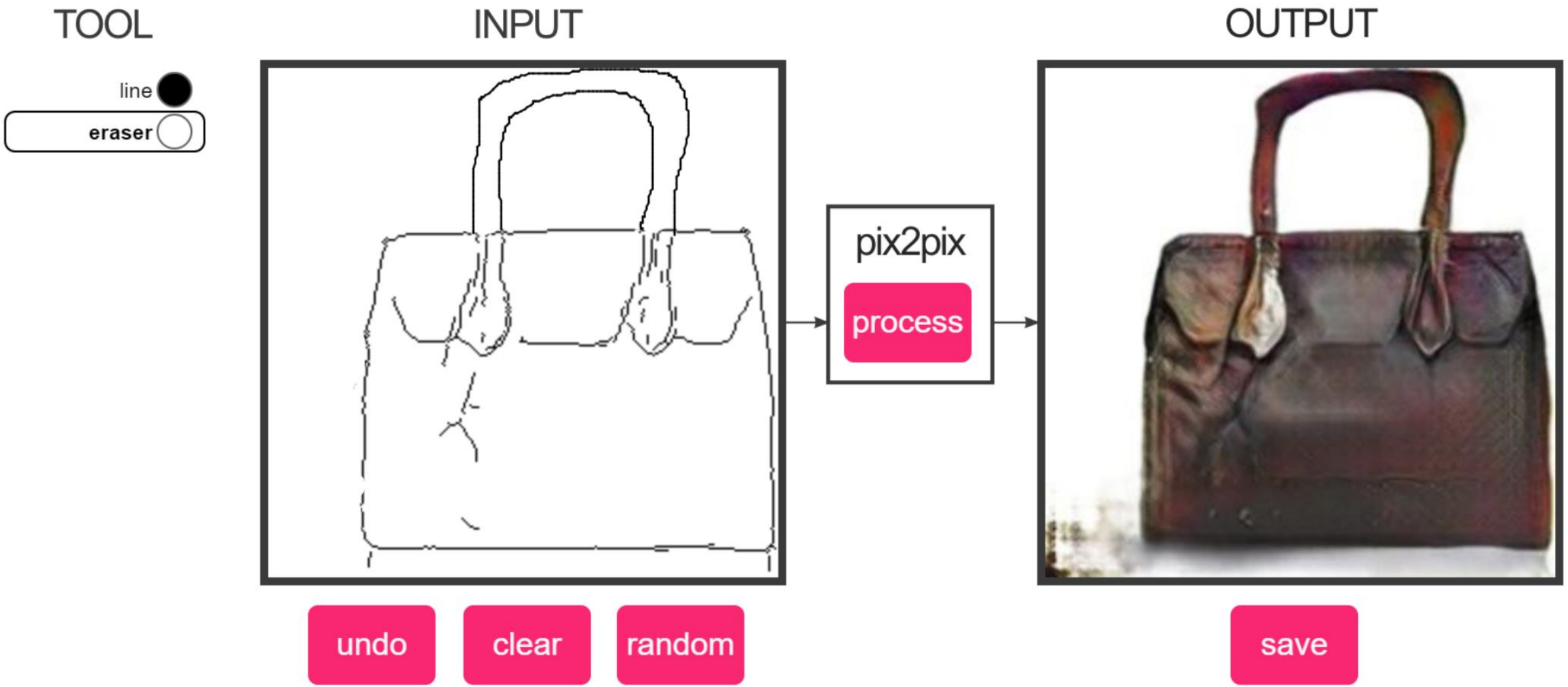
input output input

Isola, P., Zhu, J. Y., Zhou, T., & Efros, A. A. (2016). Image-to-image translation with conditional adversarial networks. arXiv preprint arXiv:1611.07004.

edges2handbags



edges2handbags

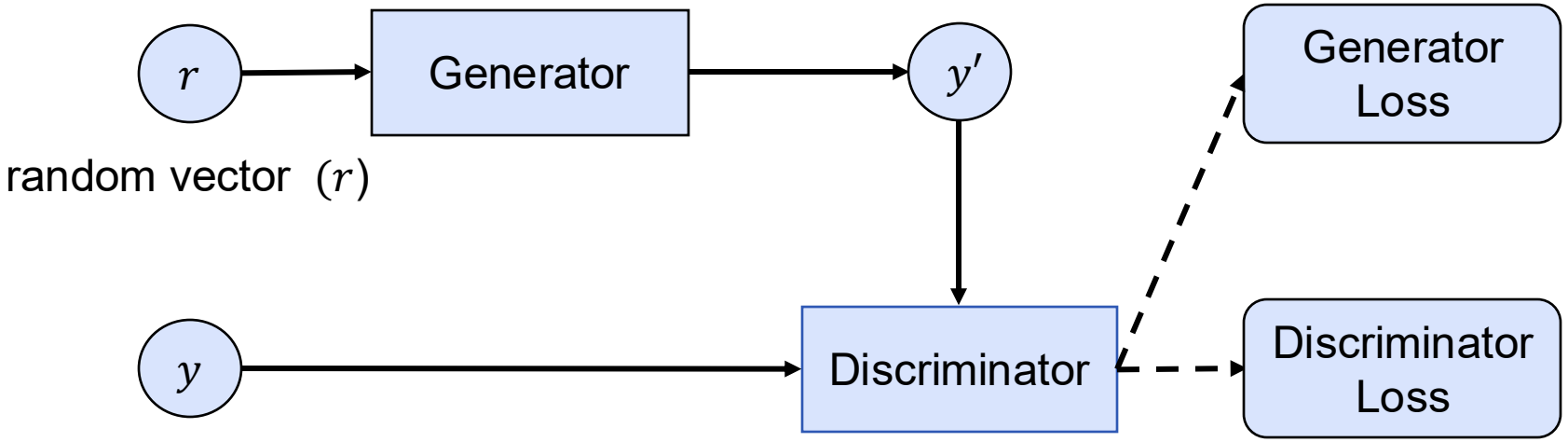


Zhu, J. Y., Krähenbühl, P., Shechtman, E., & Efros, A. A. (2016, October). Generative visual manipulation on the natural image manifold. In European Conference on Computer Vision.

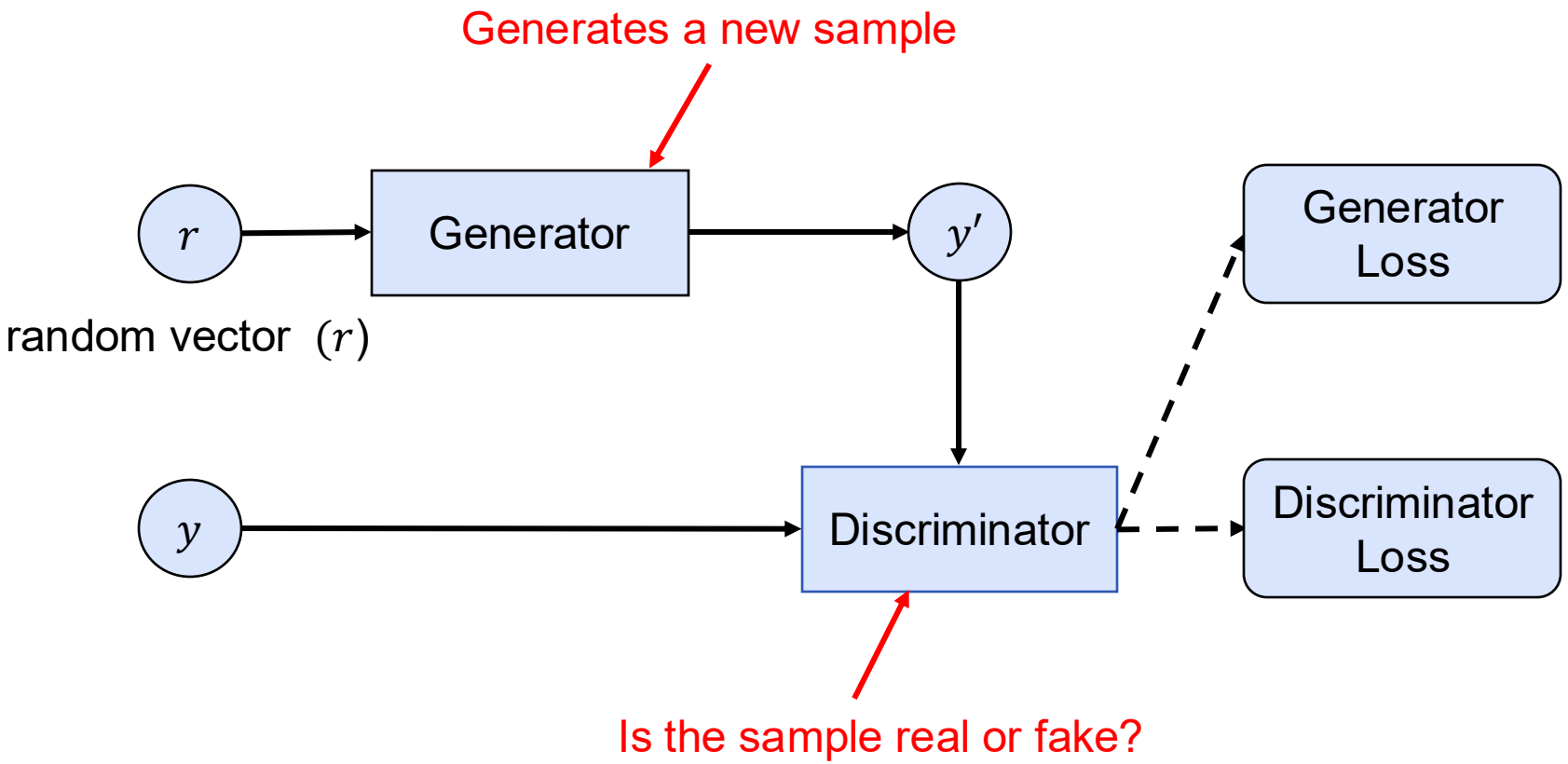
Generative Adversarial Network

- Create Data
- Style Transfer
- Increase Image Resolution (super-resolution)
- …

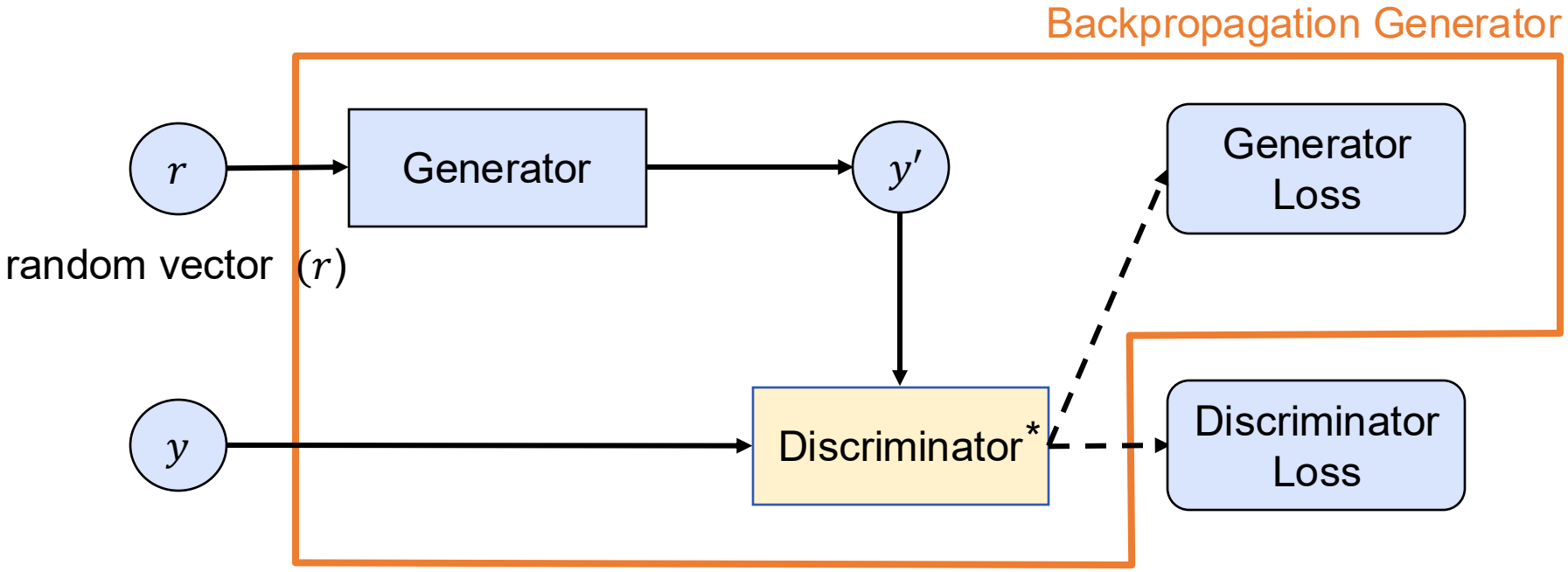
Generative Adversarial Network



Generative Adversarial Network

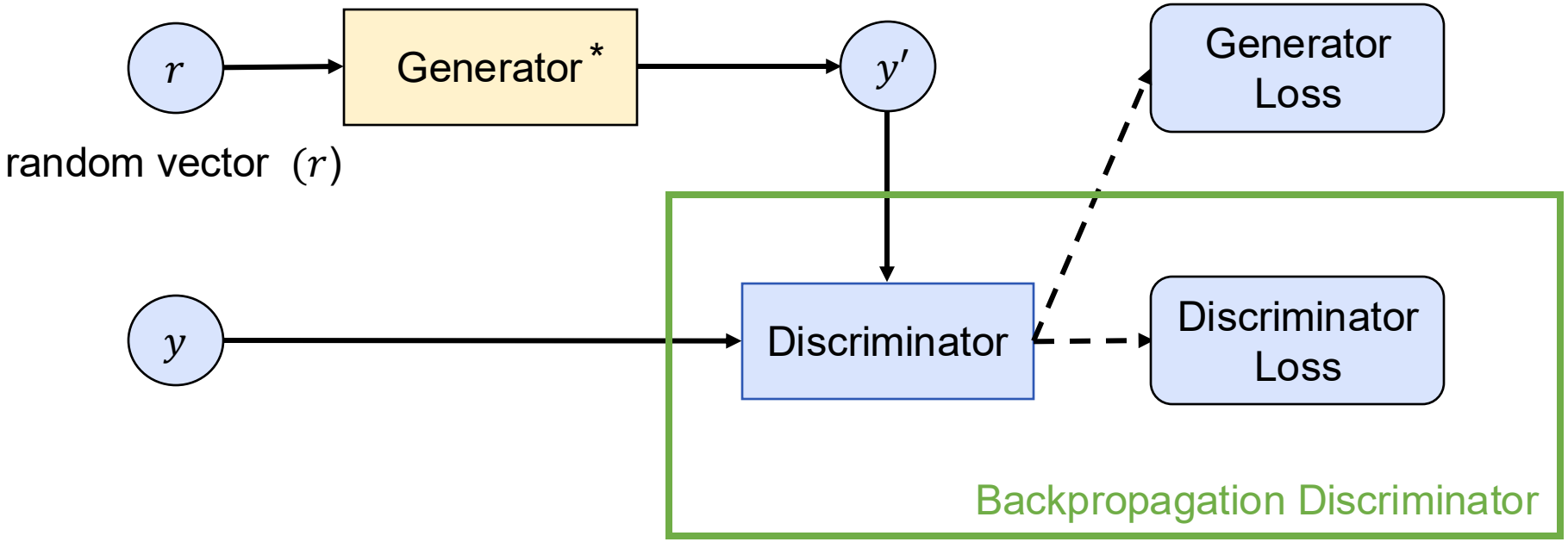


Generator Training



* Frozen model

Discriminator Training



* Frozen model

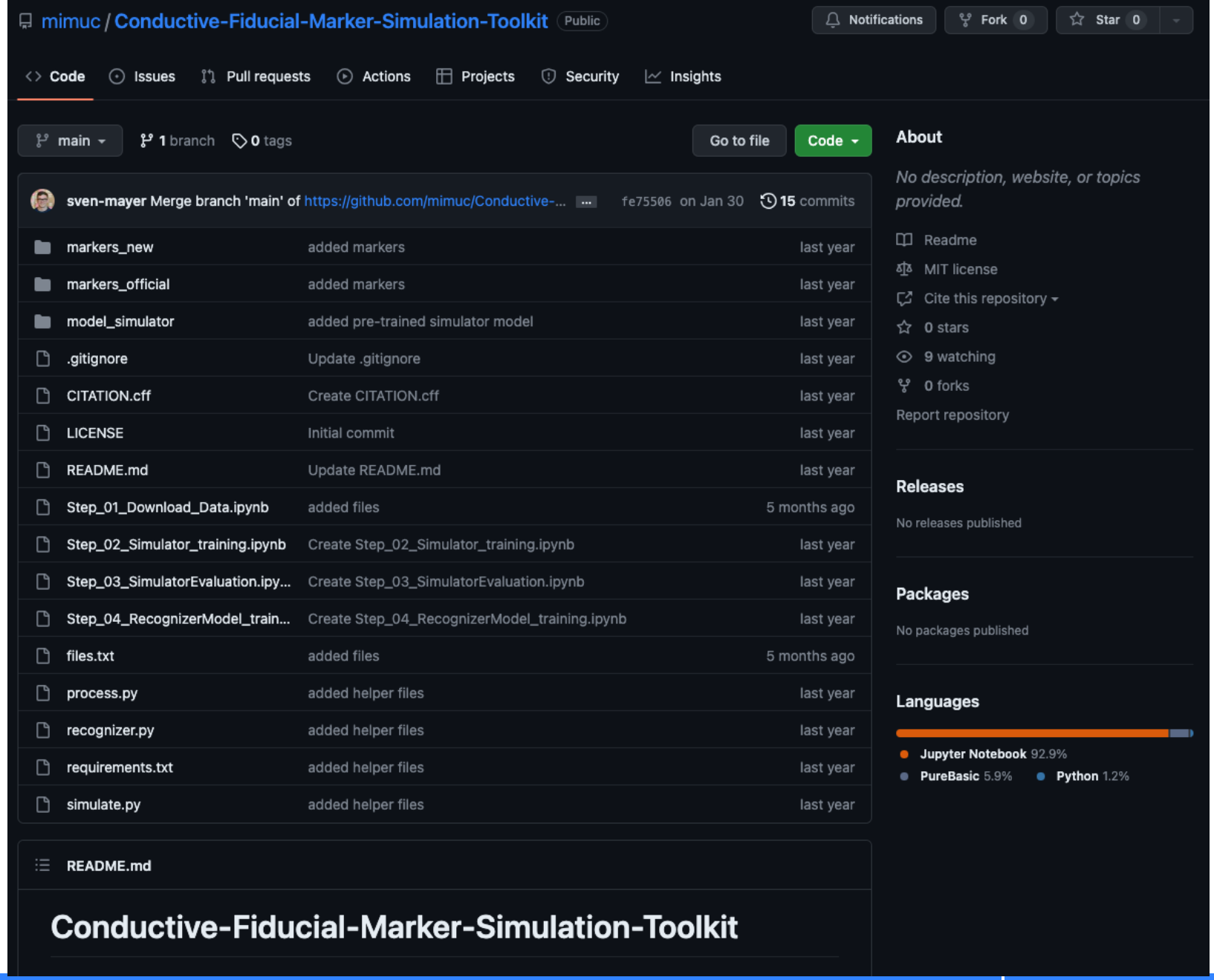
Training

During the training process the generator and the discriminator improve each other by both getting better each trying to outperform the other model.

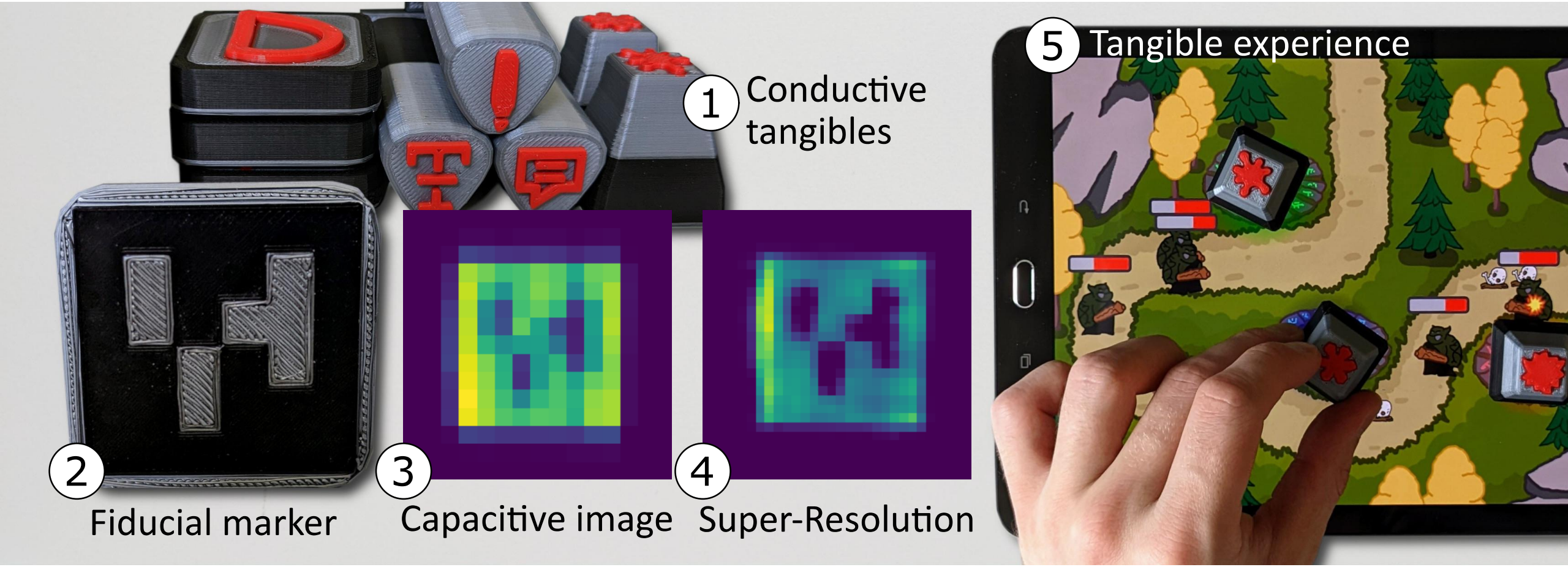
What is a good Generator?

In the end the generator is should be so good to fool the discriminator. Thus, the discriminator needs to guess fake or real (accuracy is ~.5).

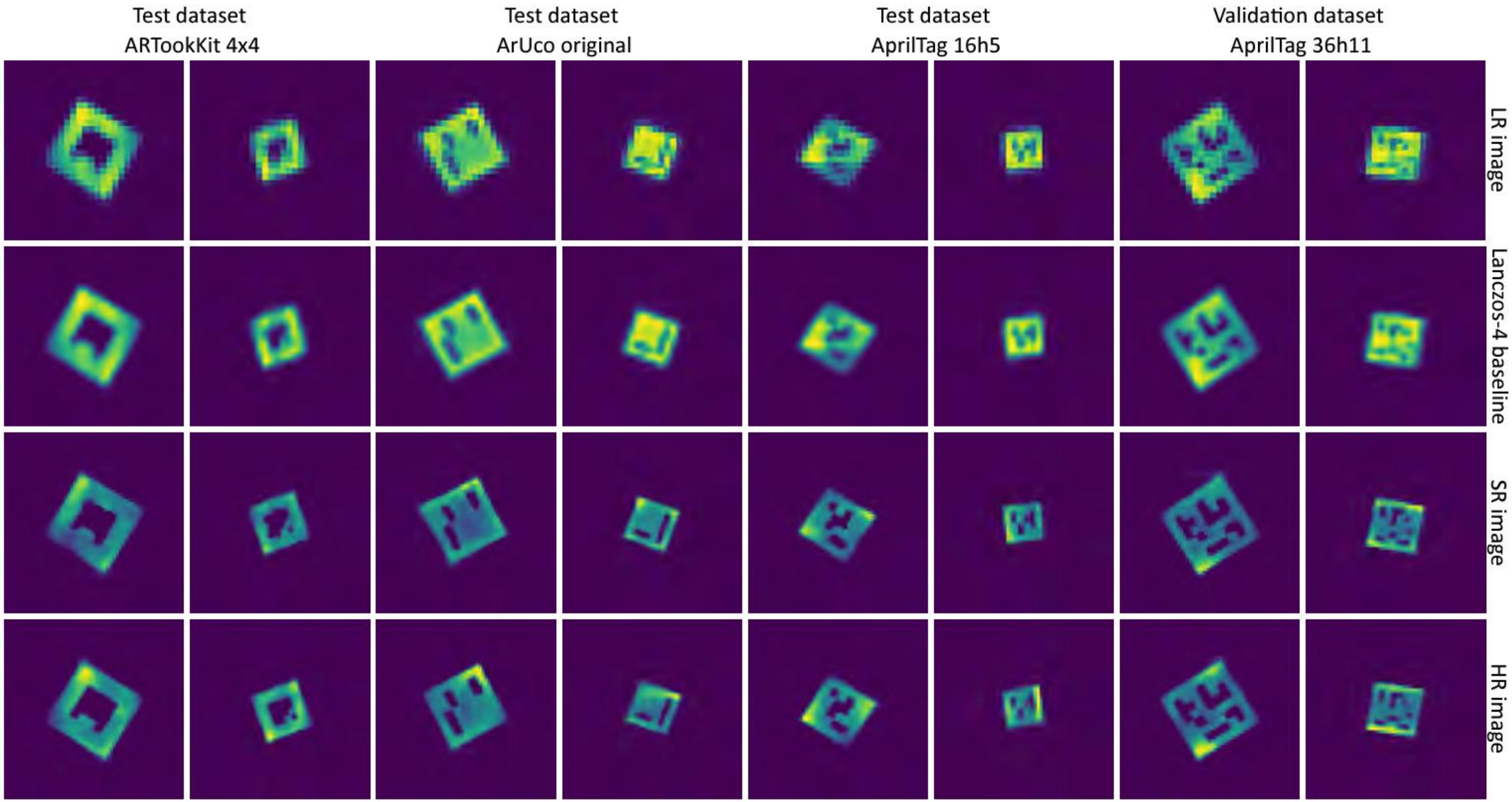
Keynote Slides


mimuc / Conductive-Fiducial-Marker-Simulation-Toolkit
Public
Notifications
Fork 0
Star 0
Code
Issues
Pull requests
Actions
Projects
Security
Insights
main
1 branch
0 tags
Go to file
Code
sven-mayer Merge branch 'main' of https://github.com/mimuc/Conductive-...
fe75506 on Jan 30
15 commits
markers_new added markers last year
markers_official added markers last year
model_simulator added pre-trained simulator model last year
.gitignore Update .gitignore last year
CITATION.cff Create CITATION.cff last year
LICENSE Initial commit last year
README.md Update README.md last year
Step_01_Download_Data.ipynb added files 5 months ago
Step_02_Simulator_training.ipynb Create Step_02_Simulator_training.ipynb last year
Step_03_SimulatorEvaluation.ipy... Create Step_03_SimulatorEvaluation.ipynb last year
Step_04_RecognizerModel_train... Create Step_04_RecognizerModel_training.ipynb last year
files.txt added files 5 months ago
process.py added helper files last year
recognizer.py added helper files last year
requirements.txt added helper files last year
simulate.py added helper files last year
README.md
Conductive-Fiducial-Marker-Simulation-Toolkit
About
No description, website, or topics provided.
Readme
MIT license
Cite this repository
0 stars
9 watching
0 forks
Report repository
Releases
No releases published
Packages
No packages published
Languages
Jupyter Notebook 92.9%
PureBasic 5.9%
Python 1.2%

Super Resolution + GAN



Marius Rusu, Sven Mayer (2023) Deep Learning Super-Resolution Network Facilitating Fiducial Tangibles on Capacitive Touchscreens. In: Proceedings of the 42nd ACM Conference on Human Factors in Computing Systems (CHI'23), Association for Computing Machinery, Hamburg, Germany.


Test dataset
ARTookKit 4x4
Test dataset
ArUco original
Test dataset
AprilTag 16h5
Validation dataset
AprilTag 36h11
LR image
Lanczos-4 baseline
SR image
HR image

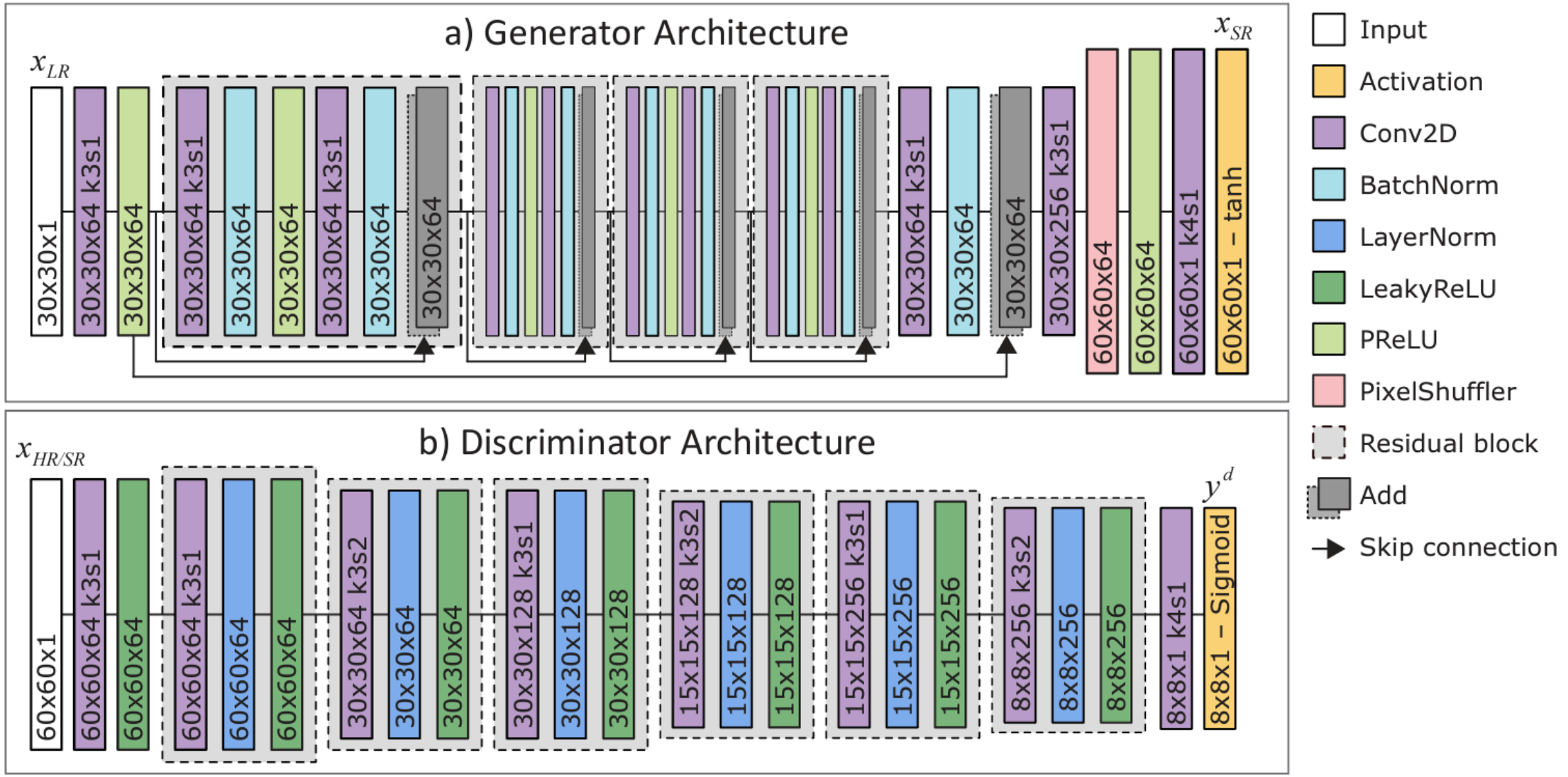
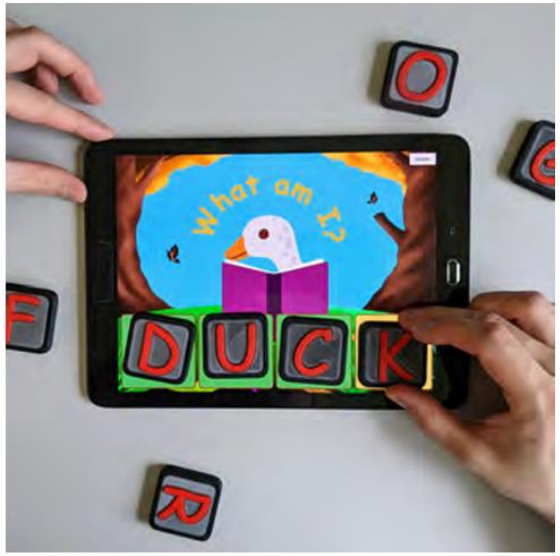


Figure 5: a) The architecture of the Generator with 1,002,433 parameters. b) The architecture of the Discriminator with 1,187,073 parameters.

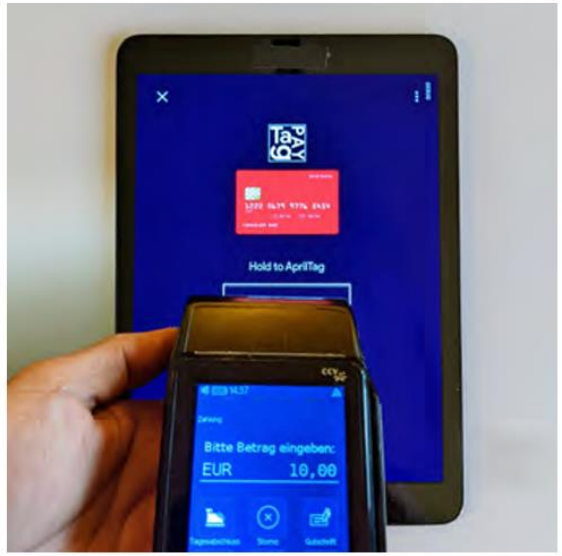
(a) Fabricated multi-material tangibles.

(b) Editing text with pen-like tangibles.

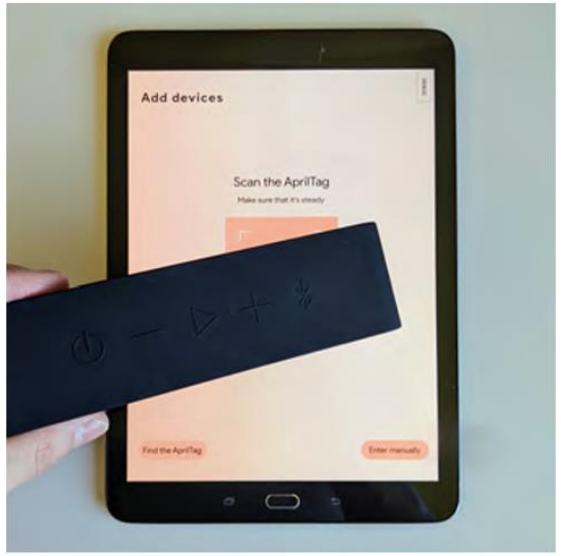
(c) Tower defense with tangible towers.



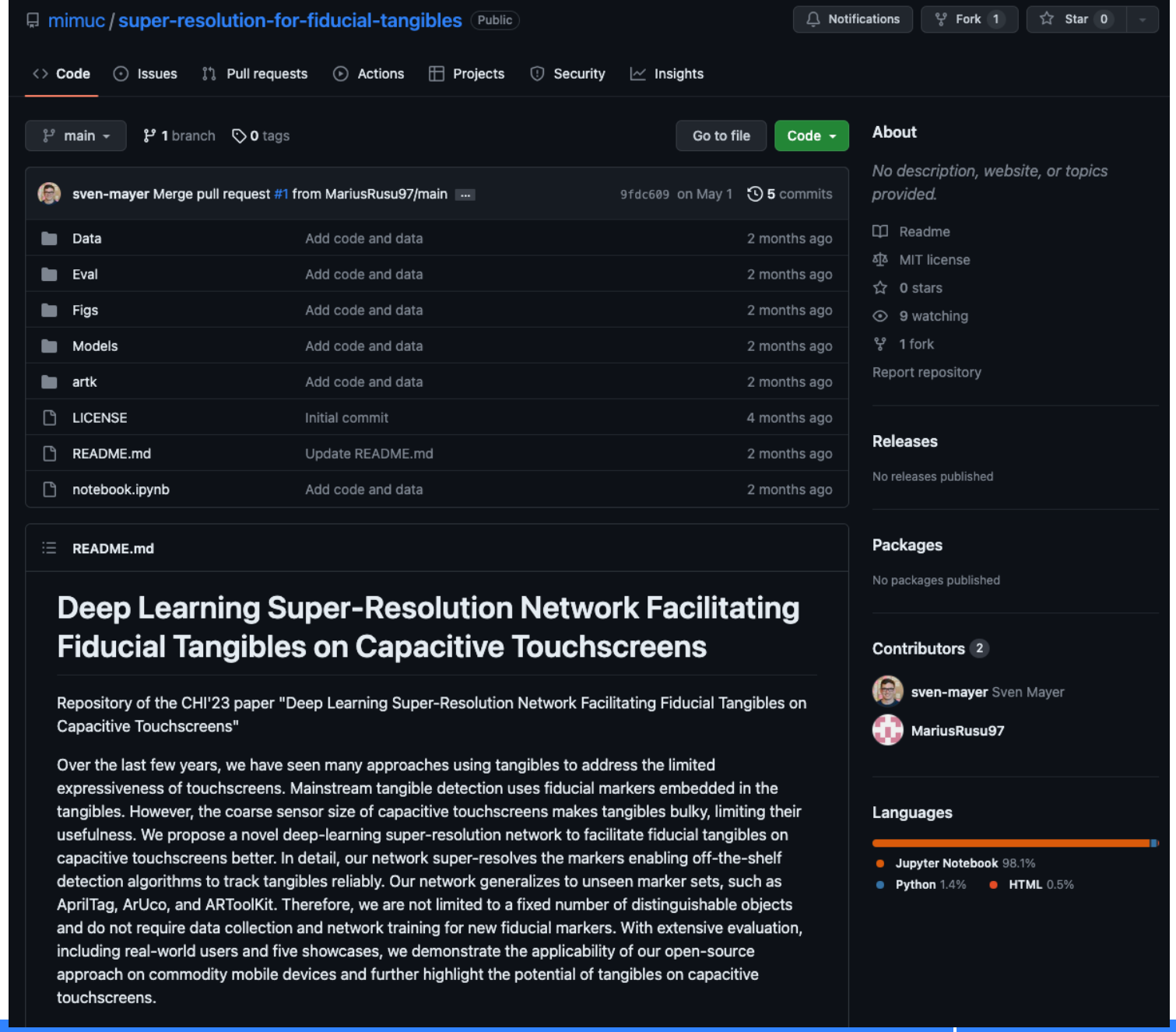
(d) Literacy learning with tangible letters.



(e) Fiducial marker to increase payment security.

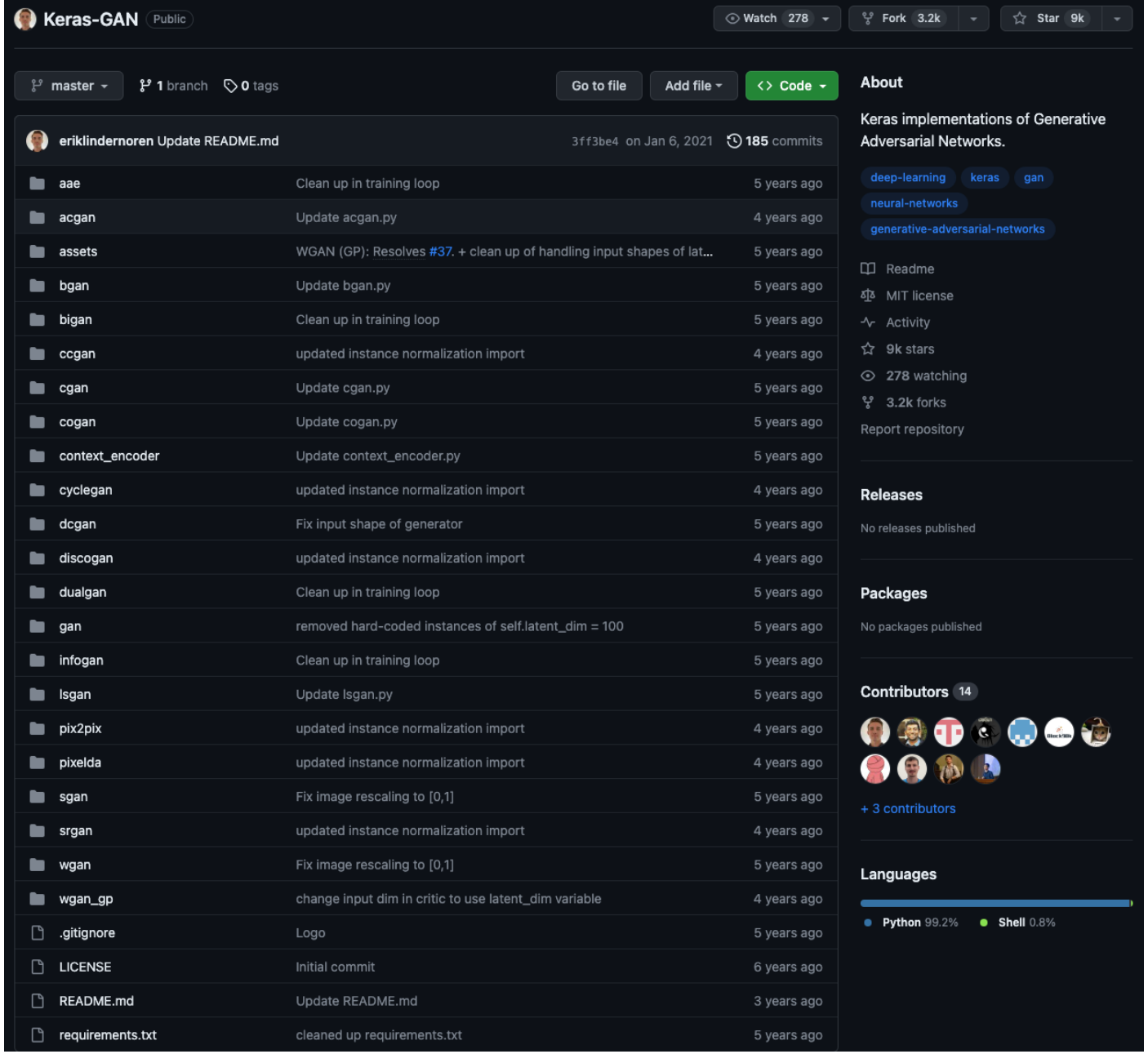


(f) Scanning a fiducial marker instead of a QR-code to connect smart devices.


mimuc / super-resolution-for-fiducial-tangibles
Public
Notifications
Fork 1
Star 0
Code
Issues
Pull requests
Actions
Projects
Security
Insights
main
1 branch
0 tags
Go to file
Code
sven-mayer Merge pull request #1 from MariusRusu97/main
9fdc609 on May 1
5 commits
Data Add code and data 2 months ago
Eval Add code and data 2 months ago
Figs Add code and data 2 months ago
Models Add code and data 2 months ago
artk Add code and data 2 months ago
LICENSE Initial commit 4 months ago
README.md Update README.md 2 months ago
notebook.ipynb Add code and data 2 months ago
README.md
Deep Learning Super-Resolution Network Facilitating Fiducial Tangibles on Capacitive Touchscreens
Repository of the CHI'23 paper "Deep Learning Super-Resolution Network Facilitating Fiducial Tangibles on Capacitive Touchscreens"
Over the last few years, we have seen many approaches using tangibles to address the limited expressiveness of touchscreens. Mainstream tangible detection uses fiducial markers embedded in the tangibles. However, the coarse sensor size of capacitive touchscreens makes tangibles bulky, limiting their usefulness. We propose a novel deep-learning super-resolution network to facilitate fiducial tangibles on capacitive touchscreens better. In detail, our network super-resolves the markers enabling off-the-shelf detection algorithms to track tangibles reliably. Our network generalizes to unseen marker sets, such as AprilTag, ArUco, and ARToolKit. Therefore, we are not limited to a fixed number of distinguishable objects and do not require data collection and network training for new fiducial markers. With extensive evaluation, including real-world users and five showcases, we demonstrate the applicability of our open-source approach on commodity mobile devices and further highlight the potential of tangibles on capacitive touchscreens.
About
No description, website, or topics provided.
Readme
MIT license
0 stars
9 watching
1 fork
Report repository
Releases
No releases published
Packages
No packages published
Contributors 2
sven-mayer Sven Mayer
MariusRusu97
Languages
Jupyter Notebook 98.1%
Python 1.4%
HTML 0.5%

GAN Architectures

That's it? No.



Conclusion

Generative Adversarial Network

- Generator
- Discriminator
- Generator tries to outperform the Discriminator to detect is if it is a real or fake image

License